

Early Fault Detection & Risk-Based Maintenance Prioritization for Wind Turbines

Using Multi-Farm SCADA Data

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Per-Timestep Risk Regression

A 24-hour SCADA window in, a risk score at **every timestep** out — not one label per window.

INPUT

10 farm-agnostic physical features (wind speed, pitch, yaw, gearbox temps, gear ratio, grid signals) — not raw sensors, which differ across farms

ARCH

2-layer GRU, hidden size 32, dropout 0.3 — chosen over LSTM/Transformer for a small dataset

OUTPUT

Sigmoid risk score $S(t) \in [0, 1]$ at each of 144 timesteps

MODEL FOOTPRINT

10,593

trainable parameters

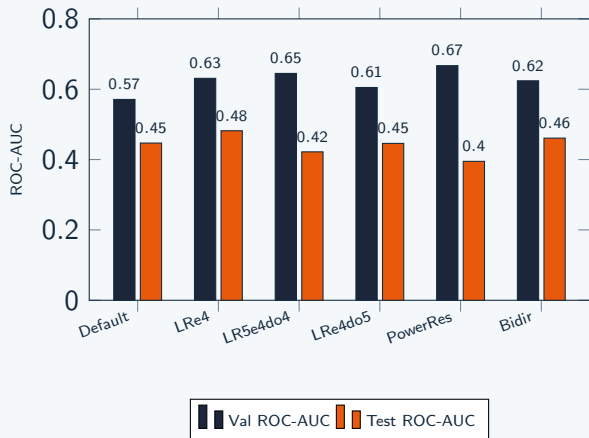
CHECKPOINT & SPEED

133.5 KB

~10ms latency/sequence

Training: Six Configurations, Tested Honestly

Selected by lowest validation loss — never by peeking at test performance



BUILT-IN, ALWAYS ON

- BCEWithLogits loss + imbalance weighting
- Masking excludes gap-filled timesteps
- Mixed precision + grad. accumulation
- Fixed seeds, fully reproducible
- Per-epoch resumable checkpointing

The Honest Finding: Val $\neq$ Test

Every one of six configurations shows the same gap: reasonable validation performance, weak held-out test performance. **Not fixed by tuning.**

VALIDATION ROC-AUC

0.57–0.67

across all 6 configs

TEST ROC-AUC

0.40–0.48

held-out turbines only

Same pattern in the classical baselines too — this is a data problem, not a GRU problem.

WHY WE PIVOTED

Farm A: only **5 turbines** →
val/test are single-turbine
estimates, high variance.

Farm C: **22 turbines** →
more stable splits, same gap
still appears.

That consistency is the evidence: the limitation is structural.

DATA & BASELINE

The Dataset: CARE-to-Compare

Fraunhofer IWES • Zenodo • CC BY-SA 4.0 • 95 labelled event datasets across 3 farms

Farm	Location	Turb.	Sensors
A	Onshore, Portugal	5	86
B	Offshore, Germany	9	257
C	Offshore, Germany (source)	22	957

EVENT DATASETS

44 + 51

anomaly events + normal controls

Farm C's larger fleet was selected as the primary training domain — more turbines means more stable train/val/test splits.

Raw Sensors Don't Match Across Farms

86, 257, and 957 raw sensor channels — a model trained on one farm's raw inputs cannot even load on another's.

THE FIX: 10 SHARED PHYSICAL FEATURES

Wind speed	Blade pitch angle	Yaw misalignment
Gearbox oil temp rise	Gearbox bearing hotspot	Gen/rotor speed ratio
Grid power factor	Current & voltage imbalance	Grid frequency deviation

Asset-disjoint splits: every turbine belongs to exactly one of train / val / test.

ALSO FOUND & FIXED

CSV delimiter bug

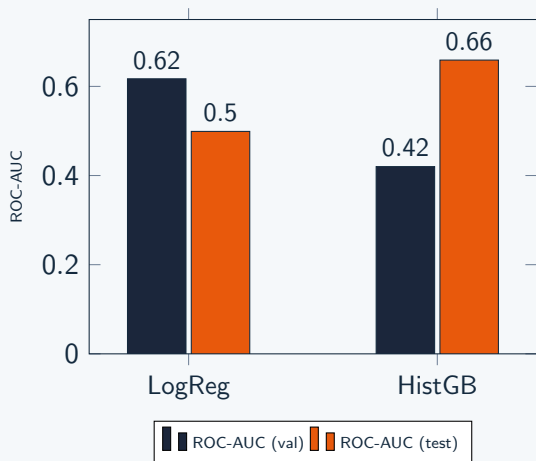
Farm C files were semicolon-delimited despite "comma_*.csv" filenames.

Threshold unit mismatch

grid_power_factor threshold was calibrated for kVA units on per-unit-scale data.

Valid fraction: 4.9% → 80.7%

The Baseline: A Bar to Actually Clear



AN EARLY WARNING SIGN

HistGradientBoosting underperforms on validation but outperforms on test — an unstable, inverted pattern.

This same instability shows up again, more severely, in the GRU results.

Balanced class/sample weighting used throughout, matching the GRU's own imbalance handling.

Evaluation Beyond Accuracy

With $\sim 4\%$ positive rate, plain accuracy is meaningless — four purpose-built metrics instead:

Lead Time

How many hours before the fault did the model first alarm?

Calibration

Does a '70% risk' prediction actually mean 70% of the time?

Cost-Sensitive

Weighted cost of missed faults vs. false alarms, at the true operating cost ratio

Per-Turbine Breakdown

Does performance hold turbine-by-turbine, or hide behind an average?

EVALUATION & DEPLOYMENT

357 Hours Lead Time — But Read the Fine Print

MEAN LEAD TIME

357.5 h

detects all 4 test fault
events

FALSE ALARM RATE

100%

alarms on every normal
window

CALIBRATION (ECE)

0.447

0 = perfect; badly over-
confident

The catch: a model that alarms almost always will trivially show a long lead time and perfect detection. This is an artifact of near-constant alarming — not genuine early, selective warning.

Per-turbine breakdown (Farm C test):

Assets 16 & 23 have a known sensor issue — F1 drops to 0.03–0.07 there vs. 0.13 on the clean turbine (56).

EVALUATION & DEPLOYMENT

Cost-Sensitive Result & Deployment Footprint

VS. ALWAYS-ALARM POLICY

62.2%

cost reduction — meaningful improvement

VS. REACTIVE-ONLY POLICY

0.11%

cost reduction — essentially no benefit (10:1 cost ratio)

MODEL SIZE & LATENCY

133.5 KB

10,593 parameters — 94ms single-sequence CPU latency

BATCHED LATENCY

10.6 ms / seq.

at batch 64 — within a 10-min SCADA interval

From Risk Score to Action

A fault-risk score alone isn't a maintenance decision — it needs context.

$$P_i = w_1 R_i + w_2 D_i - w_3 A_i + w_4 C_i$$

R — Fault Risk

Model's predicted probability, aggregated over a trailing window

D — Degradation Rate

Trend of risk over the last 6–12 hours — rising risk raises priority

A — Access Cost

Relative burden index; offshore turbines carry an illustrative 1.5× premium

C — Criticality

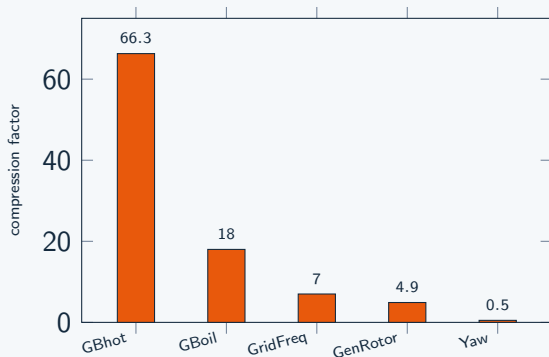
Operational importance of the asset

DECISION-SUPPORT, NOT AUTOPILOT

Given the model's calibration issues, the score's ranking is more trustworthy than its absolute magnitude. Fully unit-tested: monotonic, deterministic, reproducible.

Does It Transfer? Zero-Shot Cross-Farm Test

Train on Farm C only. No target-domain data, no adaptation, no threshold tuning on A or B.



HIGHER = MORE VARIATION
LOST

66× compression on gearbox
hotspot temperature — the
target farm's real variation is
almost entirely flattened by
the source scaler.

**Not every feature shifts the
same way** — yaw misalign-
ment is actually less com-
pressed.

Why Transfer Is Hard: Labels & Hardware

POSITIVE RATE SPREAD

2.70×

5.61% (A) vs. 2.08% (B)

GEARBOX RATIO

113/116/119

distinct, tightly-clustered per farm

THE TAKEAWAY

A fault-detection model trained on one farm should not be assumed transferable to another without auditing sensor distributions, labels, and turbine configuration first.

LABEL DERIVATION DIFFERS TOO

Farm A: operator fault log-book, explicit onset timestamps.

Farms B/C: operating-mode codes + service reports, onset inferred.

A model trained on one labelling philosophy may not transfer cleanly.

Thank You

Questions?

github.com/thedarkstringg/team-Sigmoid (tag v1.0-final)